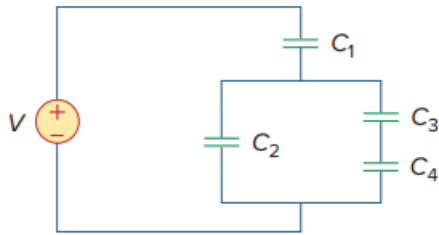


Problem

Using Fig. 6.57, design a problem that will help other students better understand how capacitors work together when connected in series and in parallel.

Fig. 6.57:



Step-by-step solution

Step 1 of 9

The series and parallel connection of capacitors is just analogous to the parallel and series connection of resistors. That is series connection of capacitor is similar to that of parallel connection of resistors and vice versa. Hence, the formula for equivalent capacitance of series connected capacitors is equal to the formula for equivalent resistance of parallel connected resistors. Similarly, the formula for equivalent capacitance of parallel connected capacitors is equal to the formula for equivalent resistance of series connected resistors

Step 2 of 9

Assume that the capacitors $C_1, C_2, C_3, \dots, C_n$ are connected series. The equivalent capacitance of this series combination is,

$$\frac{1}{C_{\text{equ}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$$

$$C_{\text{equ}} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}}$$

That is the equivalent capacitance of N series connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.

Suppose two capacitors ($N = 2$) are connected in series, then the equivalent capacitance is,

$$\frac{1}{C_{\text{equ}}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C_{\text{equ}}} = \frac{C_1 + C_2}{C_1 C_2}$$

$$C_{\text{equ}} = \frac{C_1 C_2}{C_1 + C_2} \dots\dots (1)$$

Step 3 of 9

Assume that the capacitors $C_1, C_2, C_3 + \dots + C_n$ are connected parallel. The equivalent capacitance of this parallel combination is,

$$C_{\text{equ}} = C_1 + C_2 + C_3 + \dots + C_N$$

That is the equivalent capacitance of N parallel connected capacitors is the sum of the individual capacitances.

Suppose two capacitors ($N = 2$) are connected in parallel, then the equivalent capacitance is,

$$C_{\text{equ}} = C_1 + C_2 \dots\dots (2)$$

Step 4 of 9

Refer to Figure 6.57 in the textbook.

In the circuit, capacitors C_3 and C_4 are connected in series. Find its equivalent capacitance using equation (1).

$$C_{\text{equ1}} = \frac{C_3 C_4}{C_3 + C_4} \dots\dots (3)$$

Draw the modified circuit diagram.

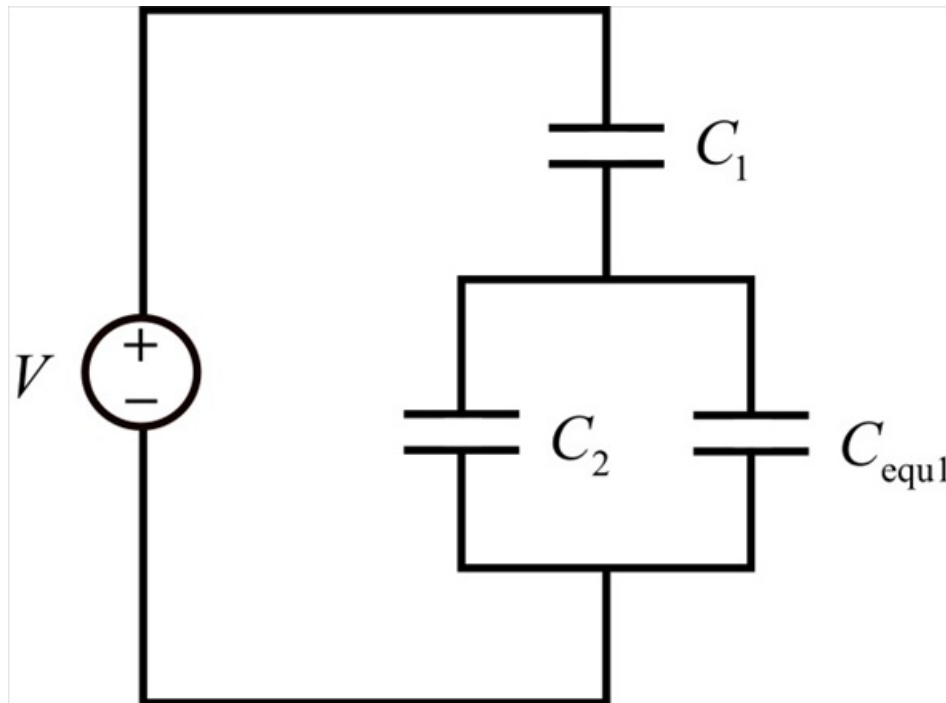


Figure 1

Step 5 of 9

In Figure 1, capacitors C_2 and C_{equ1} are connected in parallel. Find its equivalent resistance using equation (2).

$$C_{\text{equ2}} = C_2 + C_{\text{equ1}} \dots\dots (4)$$

Draw the modified circuit diagram.

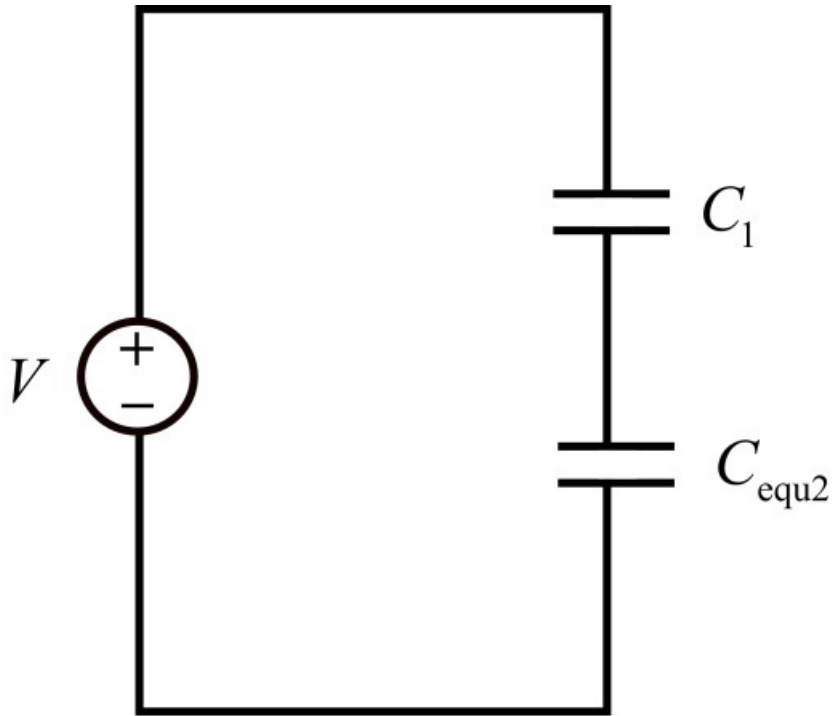


Figure 2

Step 6 of 9

In Figure 1, capacitors C_1 and C_{equ2} are connected in series. Find its equivalent resistance using equation (1).

$$C_{\text{equ}} = \frac{C_1 C_{\text{equ2}}}{C_1 + C_{\text{equ2}}} \dots\dots (5)$$

Find the value of C_{equ2} using equation (4) by substituting known value of C_{equ1} .

$$\begin{aligned} C_{\text{equ2}} &= C_2 + \frac{C_3 C_4}{C_3 + C_4} \\ &= \frac{C_2 (C_3 + C_4) + C_3 C_4}{C_3 + C_4} \\ &= \frac{C_2 C_3 + C_2 C_4 + C_3 C_4}{C_3 + C_4} \end{aligned}$$

Step 7 of 9

Substitute the value of C_{equ2} in equation (5) to find the equivalent capacitance of the circuit.

$$C_{\text{equ}} = \frac{C_1 C_{\text{equ2}}}{C_1 + C_{\text{equ2}}}$$

$$= \frac{C_1 \left(\frac{C_2 C_3 + C_2 C_4 + C_3 C_4}{C_3 + C_4} \right)}{C_1 + \frac{C_2 C_3 + C_2 C_4 + C_3 C_4}{C_3 + C_4}}$$

$$C_{\text{equ}} = \frac{C_1 C_2 C_3 + C_1 C_2 C_4 + C_1 C_3 C_4}{C_1 C_3 + C_1 C_4 + C_2 C_3 + C_2 C_4 + C_3 C_4} \dots\dots (6)$$

Therefore, the expression for the equivalent capacitance, C_{equ} of the circuit is

$$\boxed{\frac{C_1 C_2 C_3 + C_1 C_2 C_4 + C_1 C_3 C_4}{C_1 C_3 + C_1 C_4 + C_2 C_3 + C_2 C_4 + C_3 C_4}}.$$

Step 8 of 9

Assume that the capacitors in the Figure 6.57 have following values:

$$C_1 = C_2 = C_3 = C_4 = 10 \mu\text{F}$$

Step 1:

Find the value of C_{equ1} using equation (3).

$$C_{\text{equ1}} = \frac{C_3 C_4}{C_3 + C_4}$$

$$= \frac{(10 \mu\text{F})(10 \mu\text{F})}{10 \mu\text{F} + 10 \mu\text{F}}$$

$$= 5 \mu\text{F}$$

Step 2:

Find the value of C_{equ2} using equation (4).

$$C_{\text{equ2}} = C_2 + C_{\text{equ1}}$$

$$= 10 \mu\text{F} + 5 \mu\text{F}$$

$$= 15 \mu\text{F}$$

Step 3:

Find the value of C_{equ} using equation (5).

$$C_{\text{equ}} = \frac{C_1 C_{\text{equ2}}}{C_1 + C_{\text{equ2}}}$$

$$= \frac{(10 \mu\text{F})(15 \mu\text{F})}{10 \mu\text{F} + 15 \mu\text{F}}$$

$$C_{\text{equ}} = 6 \mu\text{F}$$

Step 9 of 9

Determine the equivalent resistance using equation (6) directly.

$$\begin{aligned}
 C_{\text{equ}} &= \frac{C_1 C_2 C_3 + C_1 C_2 C_4 + C_1 C_3 C_4}{C_1 C_3 + C_1 C_4 + C_2 C_3 + C_2 C_4 + C_3 C_4} \\
 &= \frac{(10 \mu\text{F})(10 \mu\text{F})(10 \mu\text{F}) + (10 \mu\text{F})(10 \mu\text{F})(10 \mu\text{F}) + (10 \mu\text{F})(10 \mu\text{F})(10 \mu\text{F})}{\left\{ (10 \mu\text{F})(10 \mu\text{F}) + (10 \mu\text{F})(10 \mu\text{F}) + (10 \mu\text{F})(10 \mu\text{F}) + \right.} \\
 &\quad \left. (10 \mu\text{F})(10 \mu\text{F}) + (10 \mu\text{F})(10 \mu\text{F}) \right\}} \\
 &= 6 \mu\text{F}
 \end{aligned}$$

Therefore, the equivalent capacitance when all capacitor values are $10 \mu\text{F}$ is $\boxed{6 \mu\text{F}}$.